Assignment No:2

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Dataset Name : Weather dataset

Problem Statement:

A dataset representing weekly weather conditions is provided. The dataset contains information about different weather parameters recorded over seven days, including:

Day of the week

Temperature (°C)

Humidity (%)

Wind Speed (km/h)

Rainfall (mm)

The objective of this problem is to perform data analysis and manipulation using Pandas and NumPy libraries in Python.

Step 1:import libraries

import numpy as np

import pandas as pd

step2:Create sample weather dataset

data = {
"Day": ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"],
"Temperature": [30, 32, 35, 33, 31, 29, 28],
"Humidity": [65, 70, 75, 72, 68, 66, 64],
"WindSpeed": [10, 12, 9, 11, 10, 8, 7],

"Rainfall": [0, 5, 10, 2, 0, 0, 0]
}

df = pd.DataFrame(data)

print(df)

output:

```
   Day  Temperature  Humidity  WindSpeed  Rainfall
0  Mon           30        65         10         0
1  Tue           32        70         12         5
2  Wed           35        75          9        10
3  Thu           33        72         11         2
4  Fri           31        68         10         0
5  Sat           29        66          8         0
6  Sun           28        64          7         0
```

Step 2: 10 Pandas Operations

1. Display first rows

print(df.head())

output:

```
   Day  Temperature  Humidity  WindSpeed  Rainfall
0  Mon           30        65         10         0
1  Tue           32        70         12         5
2  Wed           35        75          9        10
3  Thu           33        72         11         2
4  Fri           31        68         10         0
```

2. Dataset info

print(df.info())

output:

```
RangeIndex: 7 entries, 0 to 6
Data columns (total 5 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Day          7 non-null      str
 1   Temperature  7 non-null      int64
 2   Humidity     7 non-null      int64
 3   WindSpeed    7 non-null      int64
 4   Rainfall     7 non-null      int64
dtypes: int64(4), str(1)
memory usage: 412.0 bytes
None
```

3. Statistical summary

print(df.describe())

output:

```
       Temperature   Humidity  WindSpeed   Rainfall
count     7.000000   7.000000   7.000000   7.000000
mean     31.142857  68.571429   9.571429   2.428571
std       2.410295   3.994043   1.718249   3.823486
min      28.000000  64.000000   7.000000   0.000000
25%      29.500000  65.500000   8.500000   0.000000
50%      31.000000  68.000000  10.000000   0.000000
75%      32.500000  71.000000  10.500000   3.500000
max      35.000000  75.000000  12.000000  10.000000
```

4. Select specific column

print(df["Temperature"])

output:

```
0    30
1    32
2    35
3    33
4    31
5    29
6    28
Name: Temperature, dtype: int64
```

5. Filter data (Temp > 30)

print(df[df["Temperature"] > 30])

output:

```
   Day  Temperature  Humidity  WindSpeed  Rainfall
1  Tue           32        70         12         5
2  Wed           35        75          9        10
3  Thu           33        72         11         2
4  Fri           31        68         10         0
```

6. Add new column (Feels Like Temp)

df["FeelsLike"] = df["Temperature"] + 0.1 * df["Humidity"] print(df)

output:

```
   Day  Temperature  Humidity  WindSpeed  Rainfall  FeelsLike
0  Mon           30        65         10         0       36.5
1  Tue           32        70         12         5       39.0
2  Wed           35        75          9        10       42.5
3  Thu           33        72         11         2       40.2
4  Fri           31        68         10         0       37.8
5  Sat           29        66          8         0       35.6
6  Sun           28        64          7         0       34.4
```

7. Sort values

print(df.sort_values(by="Temperature"))

output:

```
   Day  Temperature  Humidity  WindSpeed  Rainfall  FeelsLike
6  Sun           28        64          7         0       34.4
5  Sat           29        66          8         0       35.6
0  Mon           30        65         10         0       36.5
4  Fri           31        68         10         0       37.8
1  Tue           32        70         12         5       39.0
3  Thu           33        72         11         2       40.2
2  Wed           35        75          9        10       42.5
```

8. Handle Missing Values(Fill NA)

df.loc[2, "Humidity"] = None

df["Humidity"] =
df["Humidity"].fillna(df["Humidity"].mean())

print(df)

output:

	Day	Temp	Humidity	WindSpeed	Rainfall
0	Mon	30	65.0	10	0
1	Tue	32	70.0	12	5
2	Wed	35	67.5	9	10
3	Thu	33	72.0	11	2
4	Fri	31	68.0	10	0
5	Sat	29	66.0	8	0
6	Sun	28	64.0	7	0

9. Drop column

```
df = df.drop(columns=["WindSpeed"])
print(df)
```

output:

	Day	Temperature	Humidity	Rainfall	FeelsLike
0	Mon	30	65	0	36.5
1	Tue	32	70	5	39.0
2	Wed	35	75	10	42.5
3	Thu	33	72	2	40.2
4	Fri	31	68	0	37.8
5	Sat	29	66	0	35.6
6	Sun	28	64	0	34.4

10. Rename column

```
df = df.rename(columns={"Temperature": "Temp"})
print(df)
```

output:

```
   Day  Temp  Humidity  Rainfall  FeelsLike
0  Mon    30        65         0       36.5
1  Tue    32        70         5       39.0
2  Wed    35        75        10       42.5
3  Thu    33        72         2       40.2
4  Fri    31        68         0       37.8
5  Sat    29        66         0       35.6
6  Sun    28        64         0       34.4
```

10 NumPy Operations

1. Convert column to NumPy array

```
temp_array = np.array(df["Temp"])
print(temp_array)
```

output:

```
[30 32 35 33 31 29 28]
```

2. Mean

```
print(np.mean(temp_array))
```

output:

```
31.142857142857142
```

3. Median

```
print(np.median(temp_array))
```

output:

```
31.0
```

4. Standard deviation

print(np.std(temp_array))

output:

```
2.231499907401901
```

5. Minimum value

print(np.min(temp_array))

output:

```
28
```

6. Maximum value

print(np.max(temp_array))

output:

```
35
```

7. Sum

print(np.sum(temp_array))

output:

```
218
```

8. Square values

print(np.square(temp_array))

output:

```
[ 900 1024 1225 1089  961  841  784]
```

9. Reshape array

print(temp_array.reshape(7, 1))

output:

```
[[30]
 [32]
 [35]
 [33]
 [31]
 [29]
 [28]]
```

10. Random noise addition

noise = np.random.normal(0, 1, len(temp_array))

print(temp_array + noise)

output:

```
[29.01644828 31.44243875 35.3357431  32.27160571 30.10964326 28.22872991
 29.83245962]
```